UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

BIOLOGY **5090/01**

Paper 1 Multiple Choice **May/June 2008**

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

8888465804

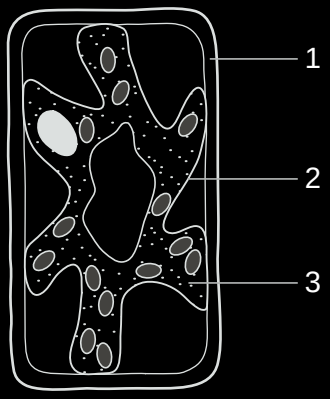
READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.



Which numbered structures are partially permeable?

A 1 and 2 only **B** 1 and 3 only **C** 1 only **D** 2 only

2 The small intestine of a person contains a low concentration of glucose produced from the digestion of starch.

Glucose is taken up by the cells of the villi.

How does the glucose move?

A by active transport against the concentration gradient

B by active transport with the concentration gradient

C by diffusion against the concentration gradient

D by diffusion with the concentration gradient

diagram 1
petri dishes viewed from the side

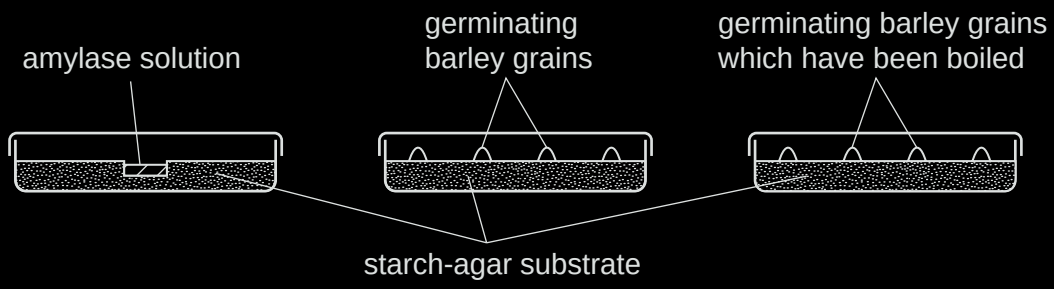
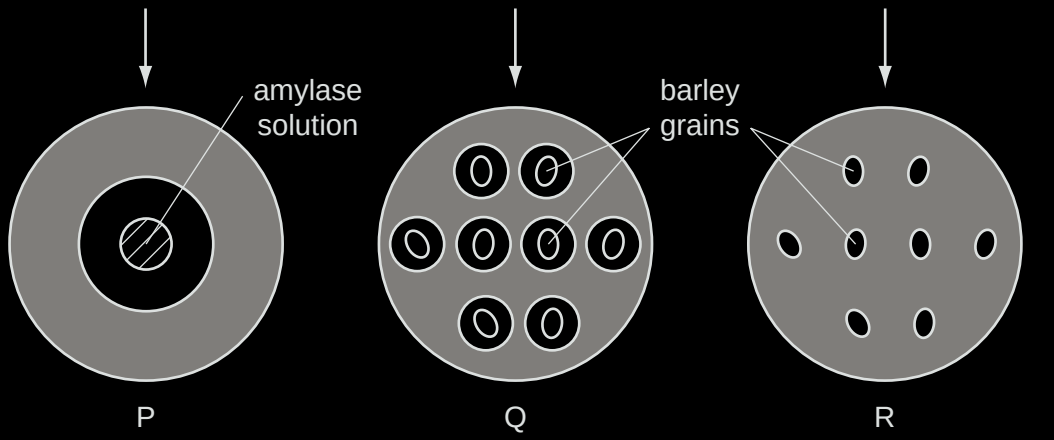


diagram 2
after 3 days, the same petri dishes viewed from above



Which is the **best** explanation of the results?

A Amylase is produced by barley grains that have been boiled.

B Amylase from barley grains is denatured when they are boiled.

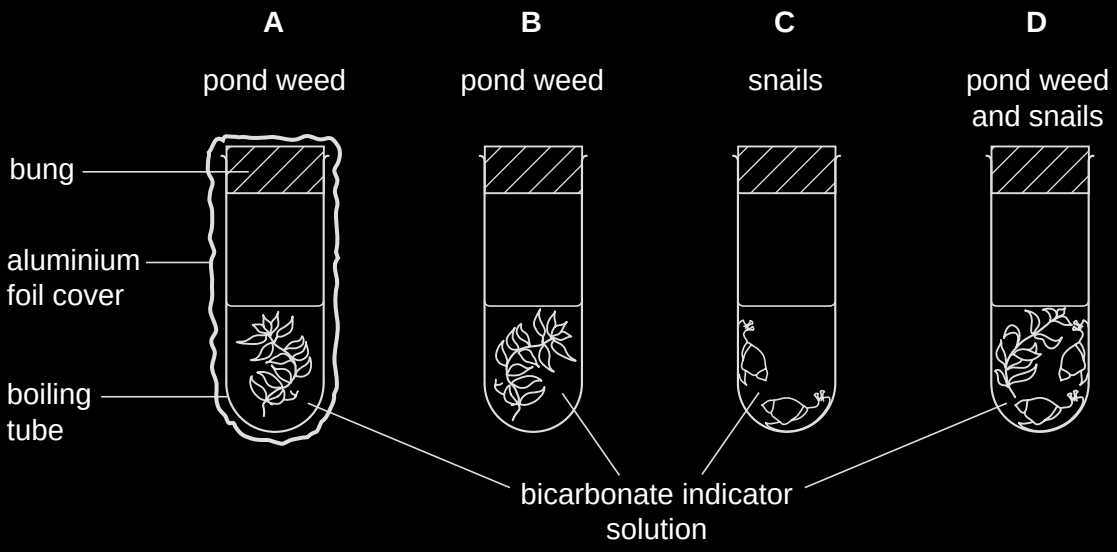
C Germinating grains prevent iodine from staining starch blue / black.

D Starch from the substrate is used by the grains as an energy source.

4 Enzyme action can be explained by the lock and key hypothesis.

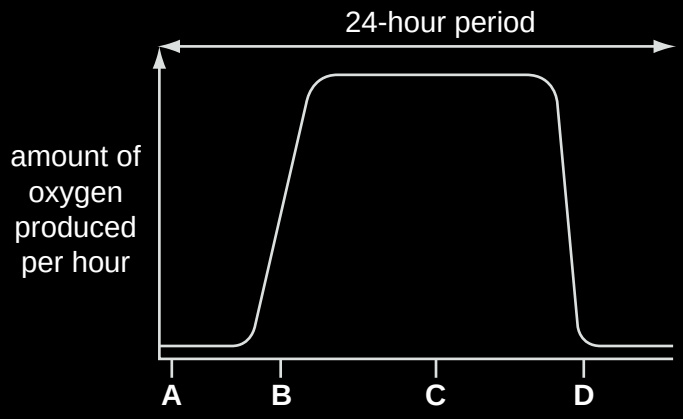
Where is the active site and which acts as the lock or key?

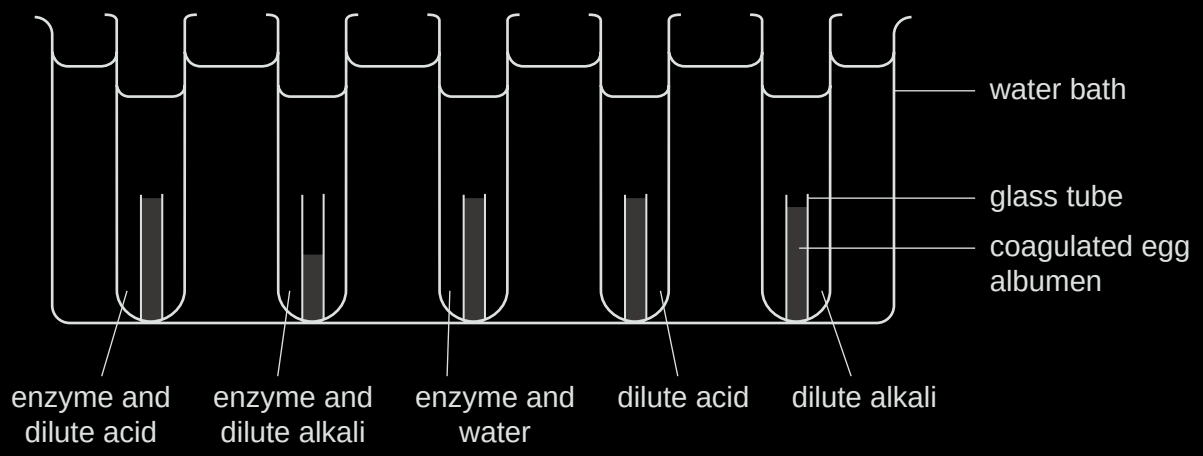
	active site	lock / key
A	on the enzyme	substrate acts as a key
B	on the enzyme	substrate acts as a lock
C	on the substrate	enzyme acts as a key
D	on the substrate	enzyme acts as a lock



7 The graph shows the amount of oxygen produced by a green plant during a 24-hour period.

Which letter represents midnight?





Where in the digestive system is the enzyme produced?

A liver

B pancreas

C salivary glands

D stomach wall

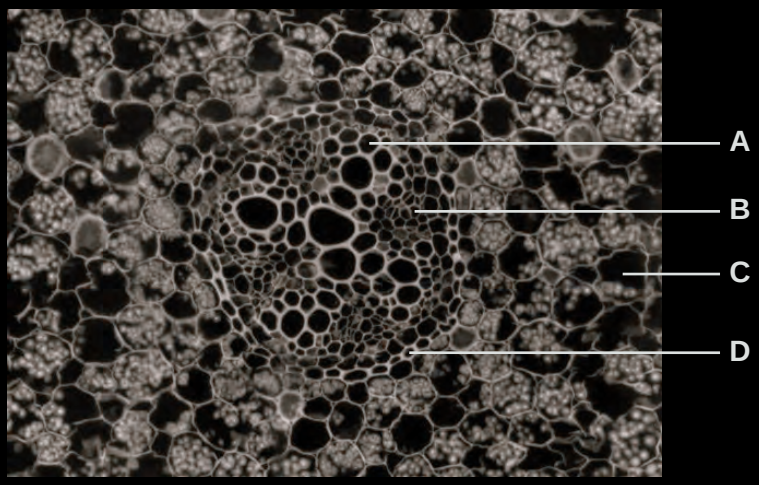
11 When a complete ring of bark is removed from the trunk of a tree, it will eventually die because this action cuts off the supply of

A mineral salts to the leaves.

B nutrients to the roots.

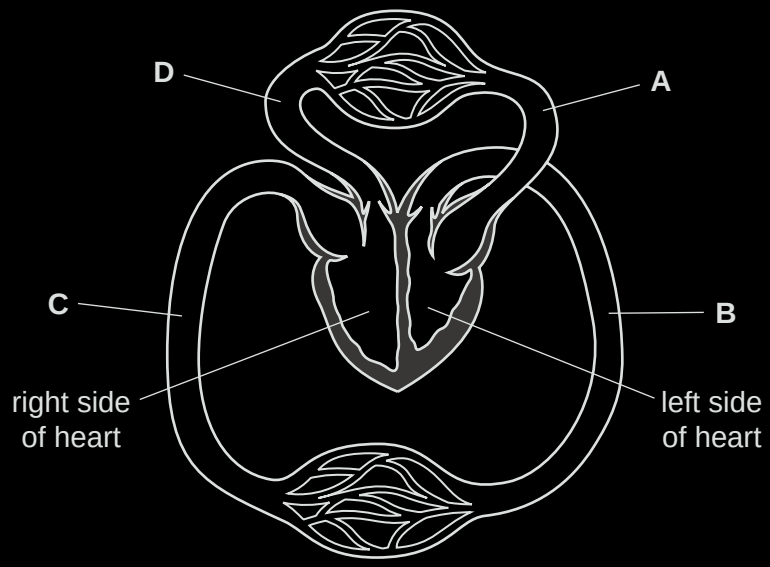
C oxygen to the roots.

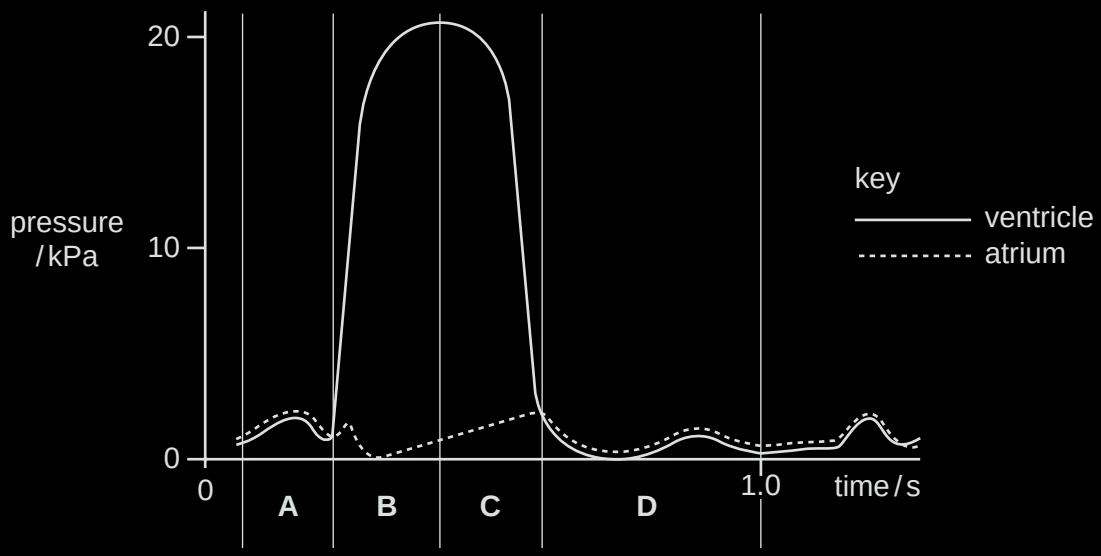
D water to the leaves.



13 The diagram represents part of the human circulatory system.

Where is the blood pressure highest?





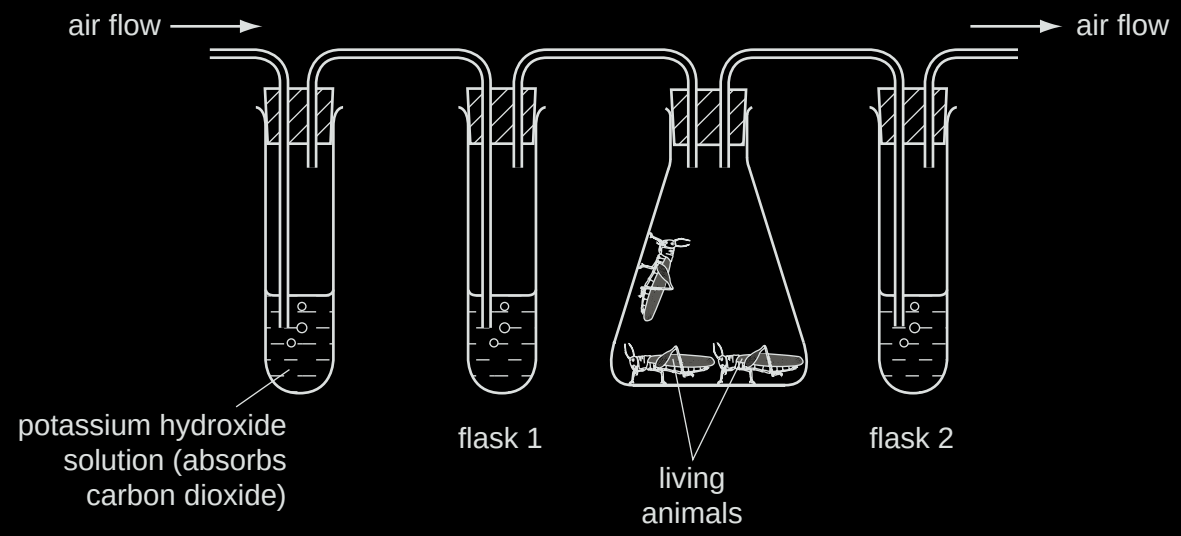
15 In the human body, blood circulating from the gut to the heart passes through the

A aorta.

B kidneys.

C liver.

D lungs.



What is the appearance of limewater in flasks 1 and 2 after a period of ten minutes?

	flask 1	flask 2
A	clear	clear
B	clear	white / cloudy
C	white / cloudy	clear
D	white / cloudy	white / cloudy

17 The composition of inhaled air and exhaled air is different.

Which analysis is correct?

	inhaled air	exhaled air
A	more carbon dioxide	less oxygen
B	less oxygen	less carbon dioxide
C	more oxygen	more carbon dioxide
D	less carbon dioxide	more oxygen

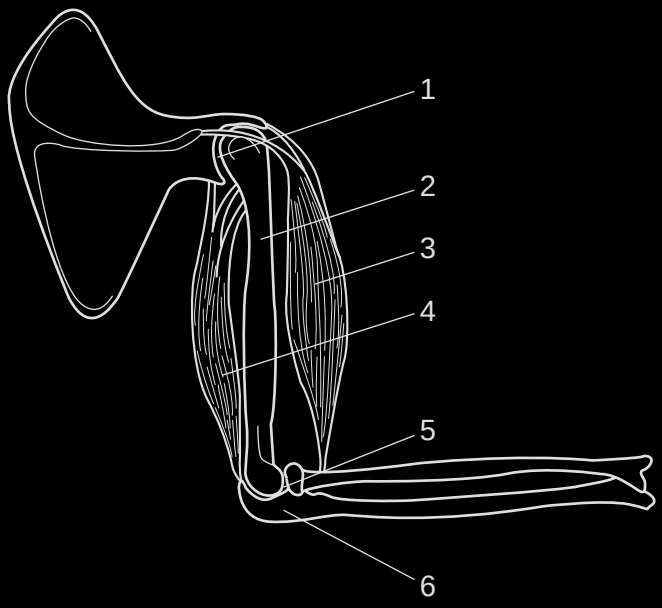
18 Which does **not** produce carbon dioxide?

A a muscle fibre

B a sensory neuron

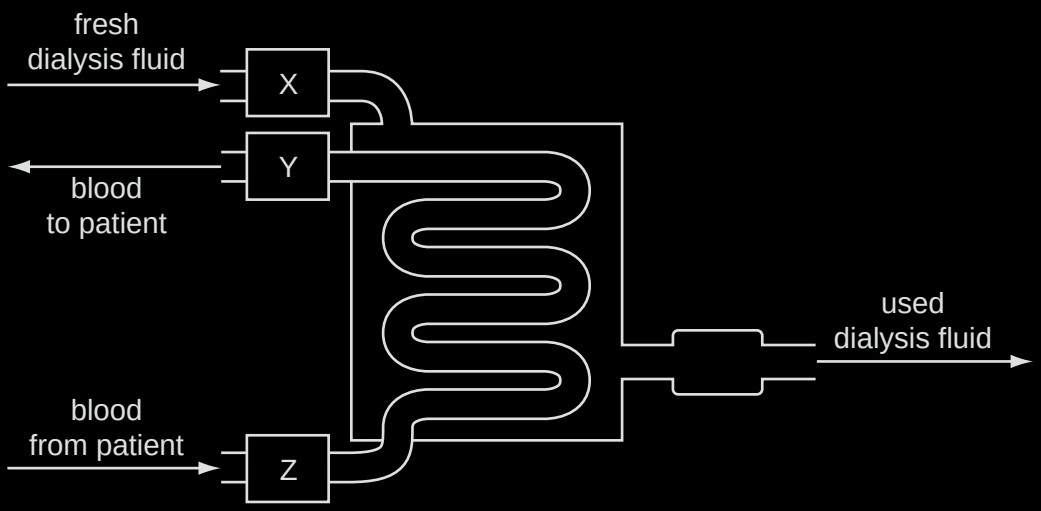
C blood

D saliva



Which lines correctly identify the biceps muscle, the humerus and a hinge joint?

	biceps	humerus	hinge joint
A	3	6	1
B	4	2	5
C	3	2	5
D	4	6	1



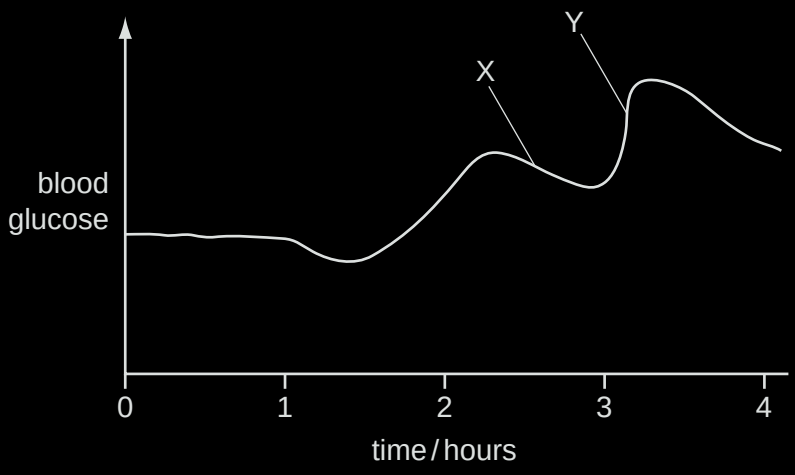
What are the parts labelled X, Y and Z?

	bubble trap	roller pump	water bath for temperature control
A	X	Y	Z
B	Y	Z	X
C	Y	X	Z
D	Z	Y	X

21 When a person is cold, nerve impulses from the hypothalamus cause the skin to reduce the rate of heat loss.

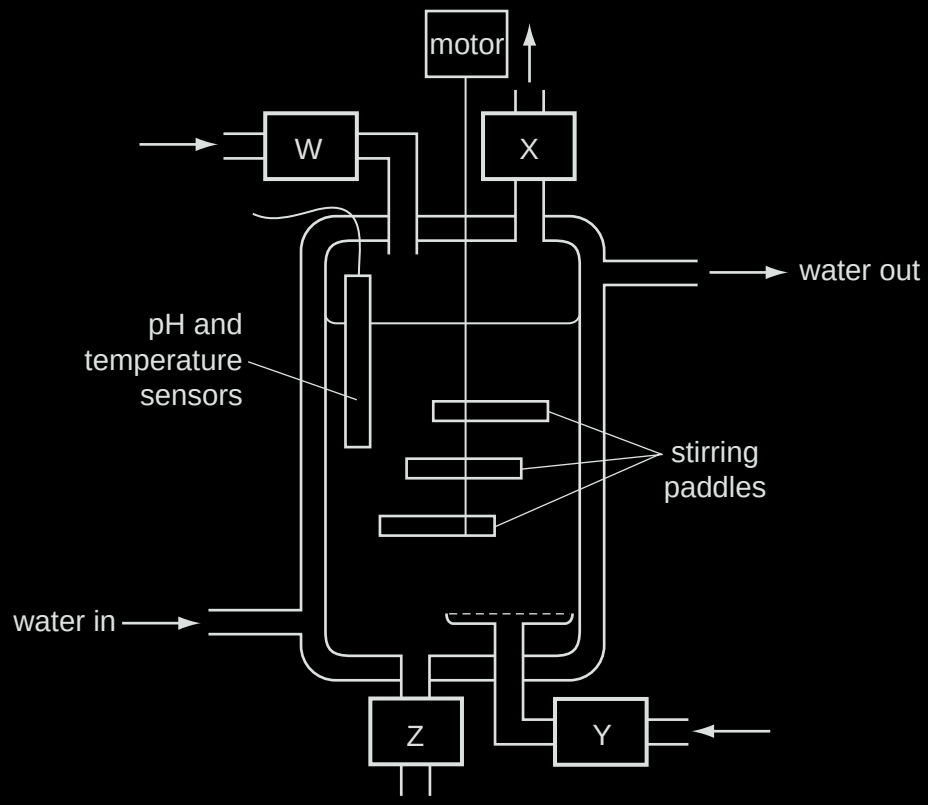
What is the effect of these nerve impulses on the hair erector muscles and on the arterioles near the skin surface?

	hair erector muscles	arterioles near skin surface
A	contract	contract
B	contract	relax
C	relax	contract
D	relax	relax



What might cause the changes at X and Y?

	X	Y
A	decreased insulin	decreased adrenaline
B	decreased insulin	increased adrenaline
C	increased adrenaline	increased insulin
D	increased insulin	increased adrenaline



What happens at W, X, Y and Z?

	air in	antibiotic out	nutrients in	waste gases out
A	W	X	Y	Z
B	W	Y	Z	X
C	Y	Z	W	X
D	Z	W	X	Z

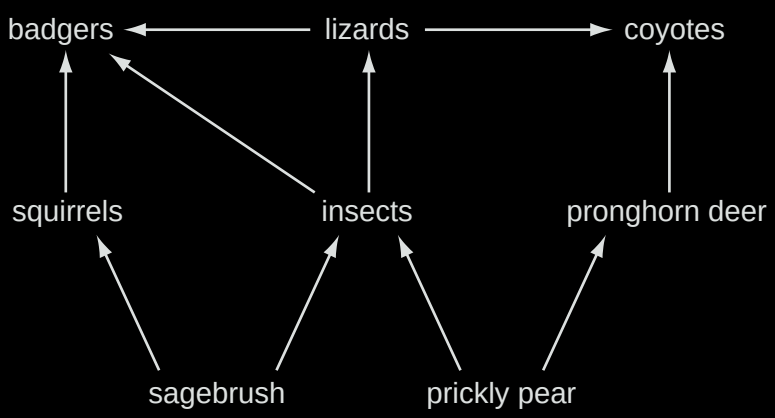
28 A decrease in the numbers of producers in a food chain will cause

A the consumers to increase in numbers.

B more animals to migrate into the area.

C the total productivity to increase.

D the total biomass to decrease.



If the population of insects decreases, which other population will decrease the most?

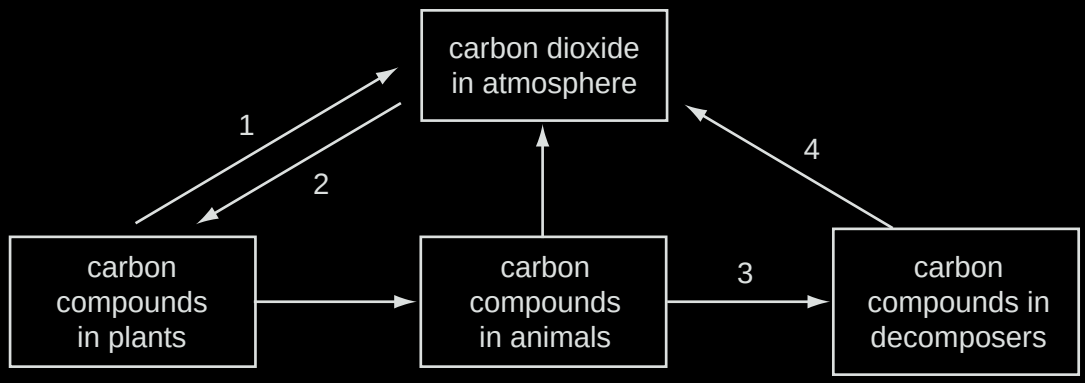
A badgers

B lizards

C sagebrush

D squirrels

30 The diagram shows part of the carbon cycle.



Which numbered arrows represent respiration?

A 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

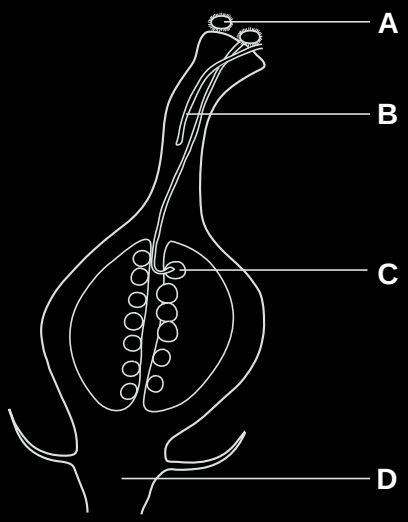
31 Which term describes the role of the mosquito in the transmission of malaria?

A parasite

B pathogen

C pest

D vector



34 In plant reproduction, the following processes occur.

1 fertilisation
2 growth of a pollen tube
3 pollination
4 seed germination

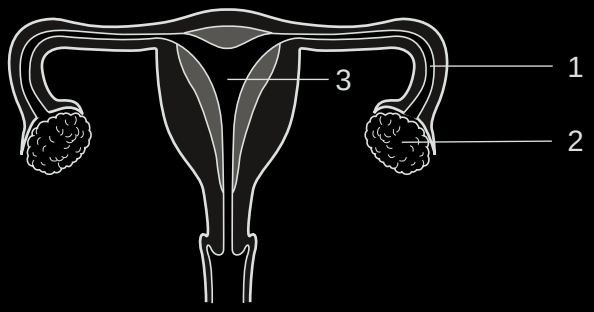
In which order do these processes take place?

A $1 \rightarrow 4 \rightarrow 2 \rightarrow 3$

B $2 \rightarrow 1 \rightarrow 3 \rightarrow 4$

C $3 \rightarrow 2 \rightarrow 1 \rightarrow 4$

D $4 \rightarrow 3 \rightarrow 1 \rightarrow 2$

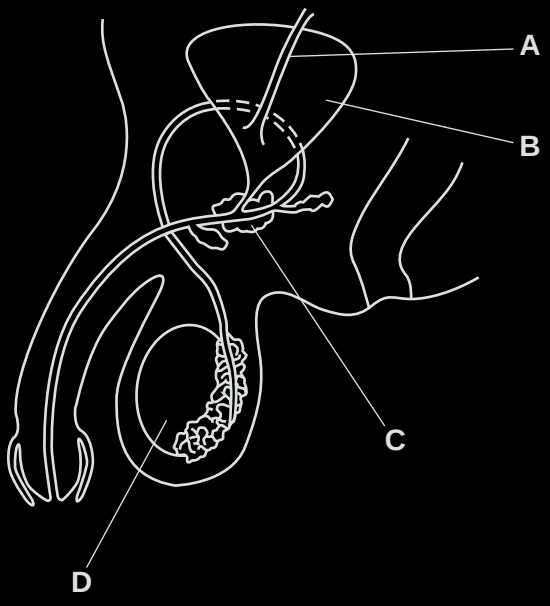


In which parts are the eggs and the zygote formed?

	eggs	zygote
A	1	2
B	1	3
C	2	1
D	2	3

36 The diagram shows the male reproductive and urinary systems.

Which structure produces the fluid part of semen?



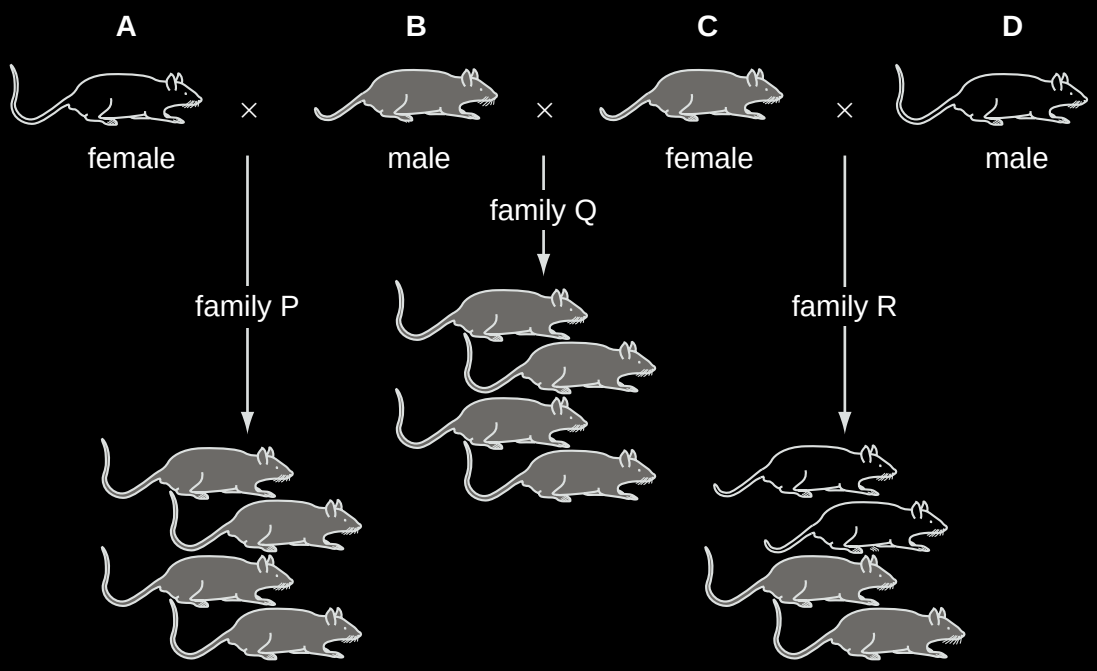
37 What causes sickle cell anaemia?

A a change in the structure of a gene

B an iron-deficient diet

C infection following a bite from a mosquito

D loss of blood following an accident



39 Two fruit flies with normal wings were mated with each other.

For every three offspring with normal wings, there is one with short wings.

What are the genotypes of the parents?

	female	male
A	Nn	Nn
B	Nn	NN
C	NN	Nn
D	Nn	nn

key

N = allele for normal wings

n = allele for short wings

40 What is a result of natural selection?

A cattle that produce large quantities of milk

B crop plants that are resistant to disease

C grapes that develop no seeds

D mosquitoes that are resistant to insecticides